PRODUCT USER MANUAL

For Arctic Ocean Sea and Ice Surface Temperature Products

SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008

SEAICE_ARC_PHY_SST-IST_L3S_NRT_011_022

Issue: 2.0

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RECORD TABLE

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1.0	2014/12/15	All	First draft version, based on on the template from the OSI-TAC Wiki	Jacob L. Høyer	Lars-Anders Breivik L. Crosnier
1.1	2015/05/01	All	Change format to fit CMEMS graphical rules	Jacob L. Høyer	L. Crosnier
1.2	2016/01/15	All	Rebranded for CMEMS	Jacob L. Høyer	B. Hackett
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1.6	2024/07/31	All	Updated information on sea ice concentration upstream data	Ioanna Karagali, Pia Englyst, Ida Lundtorp Olsen	
2.0	2025/05/09	All	New template, inclusion of the L3S 011_022 description	Ioanna Karagali, Wiebke M. Kolbe	Copernicus Marine Product Management

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
ARC	Arctic Monitoring and Forecasting Centre
AVHRR	Advanced Very High Resolution Radiometer – Optical instrument onboard NOAA satellites
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
ESA	European Space Agency
ECV	Essential Climate Variables
ENVISAT	ESA Environment Satellite
EO	Earth Observation
ERS	ESA Environment Remote Sensing satellite
EUMETSAT	European Meteorological Satellite AEurogency

EuroGOOS	European Global Operational Oceanography System
GCOS	Global Climate Observing System
GDAC	Global Data Archiving Centre
GHRSST	Group for High Resolution Sea Surface Temperature
GOOS	Global Ocean Observing System
NOAA	National Oceanic and Atmospheric Administration of the USA
SAF	EUMETSAT Satellite Application Facility
SST	Sea Surface Temperature
IST	Sea-Ice Surface Temperature

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1. INTRODUCTION

a) Summary

This document provides to CMEMS users practical information on the SI TAC level 3S and L4 Sea and Ice surface temperature products for the Arctic Ocean, which are processed at the Danish Meteorological Institute (DMI). The products are named:
SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008
SEAICE_ARC_PHY_SST-IST_L3S_NRT_011_022

The PUM describes how the products are created, the input data and methods used and the data services available to access the datasets.

The Copernicus Marine Service Arctic Sea and Sea-Ice Surface Temperature Analysis L4 product is a gap free, satellite-based, combined SST and IST analysis created by the DMIOI system (Høyer and She, 2007; Høyer et al., 2014) using L2P SST data from various sensors and high-resolution sea ice products from the Copernicus Marine Service (SEAICE_BAL_PHY_L4_MY_011_019). The corresponding L3S product is the super-collated version with all the single sensor inputs aggregated to the same grid yet without no optimal interpolation to fill-in the gaps. The products are daily fields with a 0.05 degrees resolution, and cover surface temperatures in the ocean, the sea ice and the marginal ice zone.

2. DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008 SEAICE_ARC_PHY_SST-IST_L3S_NRT_011_022
Geographical coverage	Arctic Ocean
Variables	SEAICE_ARC_PHY_SST-IST_L3S_NRT_011_022 • Sea surface temperature • SSES standard deviation • SSES bias • Sea ice fraction • Mask • Quality level • Source of satellite SST • Sum SST • Sum square SST • SST difference time • Number of SST pixels in original data SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008 • Analysed Surface Temperature • Analysis error • Sea ice fraction • Mask • Observation STD • Observation MIN • Observation MAX • Observation NUM
Product Type	Near Real Time - Analysis
Available time series	01/01/2018 (L4) 26/11/2025 (L3S)
Temporal resolution	Daily
Update Frequency	Daily

Target delivery time	12:00
Horizontal resolution	0.05 degrees
Number of vertical levels	Surface only
Format	NetCDF4

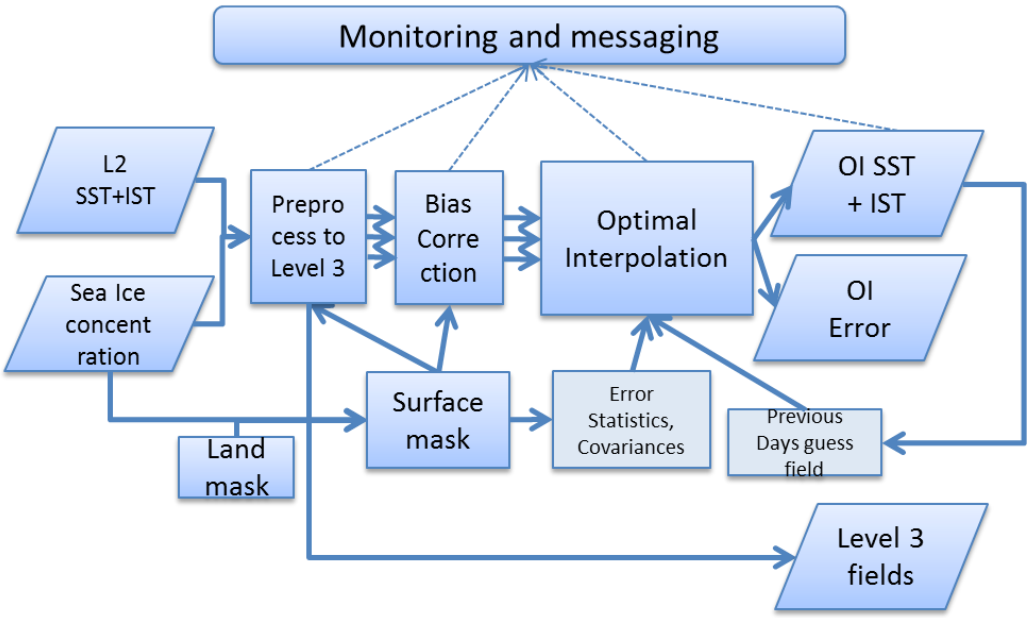
b) Production System Description

The SST and IST analysis uses the optimal interpolation method (Høyer and She, 2007; Høyer et al., 2014), which operates with anomalies from a first guess field. In the current approach, a persistence-based method is applied, which uses the previous analysis field as the first guess field. The SST and IST observations from the current day are therefore interpreted as anomalies with respect to the first guess field. The errors on the guess field are derived from the data.
Observational data for the SST and IST from satellites are Level 2 and 3U infra-red observations from the AVHRR instruments on board Metop-B (SST+IST), VIIRS_NPP (SST+IST), NOAA 20 (SST) and the SLSTR SST from Sentinel 3 A/B. The SST products are currently adjusted to the VIIRS_NPP SST observations in a dynamical way. The satellite observations are processed at DMI for high latitudes. The level 2/3U data are processed using an IST, SST or MIZ algorithm and the observations contain information about what algorithm is used and the cloud mask quality. For more information about the level 2 satellite processing, see Dybkjær et al., 2012 & 2014.

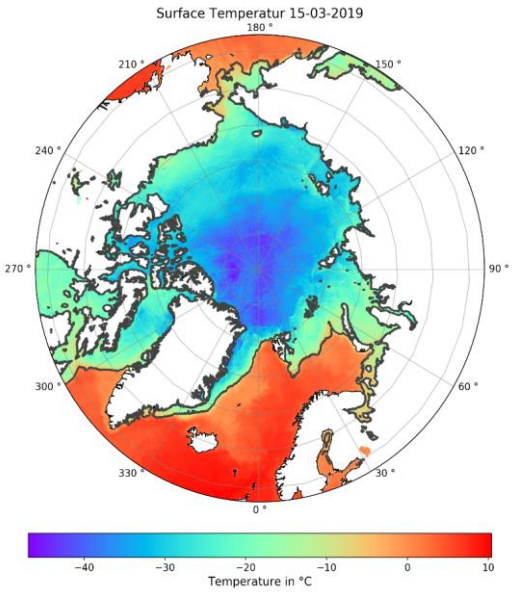
All satellite data valid for a particular day, within 24 hours from the analysis are considered. The input L2P SST data undergo various QC and processing steps to generate separate level 3 products:

- Only satellite data classified as cloud free by the PPS cloud mask are included.
- Satellite observations using an ice algorithm in regions where the sea ice concentration is 0 are discarded.
- Satellite observations processed using an SST algorithm in regions where the sea ice concentration is larger than 70% are discarded.
- Sensor specific and quality dependent biases are subtracted as a result of the validation procedure.

Sea-ice concentration data are used for the surface characterisation. The OSI-SAF sea ice concentration product OSI-401b is on a 10 km stereopolar grid has been used until November 2024 with additional filtering applied to minimize spurious see ice (in particular resulting from land spill-over) using the sea surface temperature from OSTIA (Donlon et al., 2012) product SST-GLO-SST-L4-NRT-OBSERVATIONS-010-001. In April 24, 2023 the OSI-401b product was discontinued and replaced with the OSI-401d product of the same resolution and on the same grid but with a different variable name and filtering criteria already implemented (see https://osisaf-hl.met.no/sites/osisaf-hl/files/user_manuals/osisaf_pum_ice-conc_l2-3_v1p2.pdf). In November 2024 the SIC product was substituted from OSI-401-d to OSI-408-a. Due to challenges in the coastal areas and enclosed basins caused by the larger footprint of Passive Microwave Instruments (on which OSI-401-d, OSI-401-d and OSI-408-a are based), high resolution SIC information from the CMEMS 1 km SIC product SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004 is used for the Baltic Sea, Kattegat Bay and the Skagerrak Strait.



The processing system running at DMI.

Arctic Ocean (180°W-180°E; 58°N – 90°N): 1/20°, regular grid, 640 x 7200 This product covers the Arctic Ocean north of 58°N. It is defined on a standard grid at 1/20 degree (approx. 5 km).  	
Vertical grid	Surface only
System Name	DMIOI
Model version	2.0
Horizontal Resolution	1/20°
Vertical Resolution	1 Z levels
Interpolation method	Optimal Interpolation

Assimilated observations	• AVHRR Metop-A/B SST+IST • VIIRS NPP SST+IST • VIIRS NOAA 20 SST • SLSTR S3-A/B SST Sea-ice concentration data: AMSR2 Global Sea Ice Concentration (OSI-408-a) from EUMETSAT OSISAF Baltic high resolution sea ice concentration from the Copernicus Marine Service SI TAC (SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004)

c) Dataset details

Product Lines	SEAICE_ARC_SEAICE_L4_NRT_OBSERVATIONS_011_008
Geographical coverage	Arctic
Datasets: DMI-ARC-SEAICE_TEMP-L4-NRT-OBS	
Product Type	Analysis
Update frequency	Daily
Available time series	2018/01/01 - up to real-time
Target delivery time	Daily 12pm (noon) UTC
Temporal resolution	Daily mean

Dataset : DMI-ARC-SEAICE_TEMP-L4-NRT-OBS contains the 2D daily fields: surface temperature, analysis error, sea_ice_fraction, mask
analysed_st [K] analysed sea and ice surface temperature surface_temperature
analysis_error [K] estimated error standard deviation of analysed_st
observation_std [K] level 3 standard deviation of observed surface temperature in grid point
observation_min [K] level 3 minimum observed surface temperature in grid point
observation_max [K] level 3 maximum observed surface temperature in grid point
observation_num number observations in grid point
mask land sea ice lake bit mask
sea_ice_fraction sea ice area fraction sea_ice_area_fraction

Product Lines	SEAICE_ARC_PHY_SST-IST_L3S_NRT_011_022
Geographical coverage	Arctic
Datasets: cmems_obs-si_arc_phy-temp_nrt_l3s_P1D-m	
Product Type	Analysis
Update frequency	Daily
Available time series	2025/11/01 - up to real-time
Target delivery time	Daily 12pm (noon) UTC
Temporal resolution	Daily mean

Dataset : cmems_obs-si_arc_phy-temp_nrt_l3s_P1D-m contains the 2D daily fields: surface temperature, analysis error, sea_ice_fraction, mask
analysed_st [K] analysed sea and ice surface temperature surface_temperature
analysis_error [K] estimated error standard deviation of analysed_st
observation_std [K] level 3 standard deviation of observed surface temperature in grid point
observation_min [K] level 3 minimum observed surface temperature in grid point
observation_max [K] level 3 maximum observed surface temperature in grid point
observation_num number observations in grid point
mask land sea ice lake bit mask
sea_ice_fraction sea ice area fraction sea_ice_area_fraction

d) Processing information

Product disseminating schedule :

Type of product (forecast/ hindcast)	Forecast run (base time)	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
Analysis	12 UTC	0 to 24	10:50am -> 12 noon	Every day

e) Catalogue updating

Every day the daily L3S and L4 SST/IST analysis of the previous day is delivered.

f) Time averaging

Dataset frequency	Time averaging	Forecast lead time
daily	Each daily SST/IST analysis is based on aggregated input data from the analysis day. The previous days analysis is used as a first guess field for the OI processing system.	N/A

g) Product use and impact

For applications requiring daily monitoring of the combined Sea Surface and Sea Ice Surface Temperatures and up-to-date analysis of the Arctic Ocean, there are several reasons to choose the L4 product. Some of the advantages are listed below:

- Increased spatial resolution.
- Regional, high resolution land mask gives a better representation of small islands and complex coastal regions.
- The high resolution 1 km sea ice product derived from ice charts is significantly better in this coastal region than the global Microwave-based products.
- Regional Interpolation statistics improves the results in case of missing data.
- Combined SST and IST information and the representation of the opean ocean, marginal ice zone and sea-ice covered regions provides a consistent product for the monitoring of the surface temperature that interacts with the atmosphere.

3. FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4. FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

4.1. FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
DMI-ARC-SEAICE_TEMP-L4-NRT-OBS	<yyyymmdd>120000-DMI-L4_GHRSST-Stskin-DMI_OI-ARC_IST-v02.0-fv01.0.nc	0.065
cmems_obs-si_arc_phy-temp_nrt_l3s_P1D-m	<yyyymmdd>120000-DMI-L3S_GHRSST-Stskin-DMI_OI-ARC_IST-v02.0-fv01.0.nc	0.060

4.2. STRUCTURE AND SEMANTIC OF NetCDF MAP FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5. REFERENCES

Quality Information Document
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SEAICE-QUID-011-008-022.pdf

6. ANNEX

Structure and semantic of NetCDF maps files

For dataset DMI-ARC-SEAICE_TEMP-L4-NRT-OBS

```
netcdf \20250507120000-DMI-L4_GHRSST-STskin-DMI_OI-ARC_IST-v02.0-fv01.0 {
dimensions:
        lon = 7200 ;
        lat = 640 ;
        time = 1 ;
variables:
        float lon(lon) ;
                lon:standard_name = "longitude" ;
                lon:long_name = "longitude" ;
                lon:units = "degrees_east" ;
                lon:axis = "X" ;
                lon:valid_min = -180.f ;
                lon:valid_max = 180.f ;
                lon:comment = "geographical coordinates, WGS84 projection" ;
        float lat(lat) ;
                lat:standard_name = "latitude" ;
                lat:long_name = "latitude" ;
                lat:units = "degrees_north" ;
                lat:axis = "Y" ;
                lat:valid_min = -90.f ;
                lat:valid_max = 90.f ;
                lat:comment = "geographical coordinates, WGS84 projection" ;
        int time(time) ;
                time:standard_name = "time" ;
```

```
            time:long_name = "reference time of sst field" ;
            time:units = "seconds since 1981-01-01 00:00:00" ;
            time:axis = "T" ;
            time:comment = "Nominal time of Level 4 analysis" ;
            time:calendar = "gregorian" ;
        short analysed_st(time, lat, lon) ;
            analysed_st:standard_name = "surface_temperature" ;
            analysed_st:long_name = "analysed sea and ice surface temperature" ;
            analysed_st:units = "kelvin" ;
            analysed_st:scale_factor = 0.01f ;
            analysed_st:add_offset = 273.15f ;
            analysed_st:_FillValue = -32768s ;
            analysed_st:comment = "" ;
            analysed_st:source = "Metop-A SST and IST products from DMI" ;
            analysed_st:valid_min = -6000s ;
            analysed_st:valid_max = 4500s ;
        short analysis_error(time, lat, lon) ;
            analysis_error:long_name = "estimated error standard deviation of analysed_st"
;
            analysis_error:units = "kelvin" ;
            analysis_error:scale_factor = 0.01f ;
            analysis_error:add_offset = 0.f ;
            analysis_error:_FillValue = -32768s ;
            analysis_error:comment = "" ;
            analysis_error:valid_min = 0s ;
            analysis_error:valid_max = 32767s ;
        short observation_std(time, lat, lon) ;
            observation_std:long_name = "level 3 standard deviation of observed surface
temperature in grid point" ;
```

```
                observation_std:units = "kelvin" ;
                observation_std:scale_factor = 0.01f ;
                observation_std:add_offset = 0.f ;
                observation_std:_FillValue = -32768s ;
                observation_std:comment = "" ;
                observation_std:valid_min = 0s ;
                observation_std:valid_max = 32767s ;
        short observation_min(time, lat, lon) ;
                observation_min:long_name = "level 3 minimum observed surface temperature
in grid point" ;
                observation_min:units = "kelvin" ;
                observation_min:scale_factor = 0.01f ;
                observation_min:add_offset = 273.15f ;
                observation_min:_FillValue = -32768s ;
                observation_min:comment = "" ;
                observation_min:valid_min = -6000s ;
                observation_min:valid_max = 4500s ;
        short observation_max(time, lat, lon) ;
                observation_max:long_name = "level 3 maximum observed surface
temperature in grid point" ;
                observation_max:units = "kelvin" ;
                observation_max:scale_factor = 0.01f ;
                observation_max:add_offset = 273.15f ;
                observation_max:_FillValue = -32768s ;
                observation_max:comment = "" ;
                observation_max:valid_min = -6000s ;
                observation_max:valid_max = 4500s ;
        short observation_num(time, lat, lon) ;
                observation_num:long_name = "number observations in grid point" ;
```

```
                observation_num:units = "1" ;
                observation_num:scale_factor = 1.f ;
                observation_num:add_offset = 0.f ;
                observation_num:_FillValue = -32768s ;
                observation_num:comment = "" ;
                observation_num:valid_min = 0s ;
                observation_num:valid_max = 32767s ;
        byte mask(time, lat, lon) ;
                mask:long_name = "land sea ice lake bit mask" ;
                mask:_FillValue = -128b ;
                mask:comment = "Mask can be used to further filter the data" ;
                mask:flag_meanings = "water land optional_lake_surface sea_ice
optional_river_surface" ;
                mask:source = "NAVOCEANO_landmask_v1.0, OSI-SAF sea ice
concentration" ;
                mask:flag_masks = 1b, 2b, 4b, 8b, 16b ;
                mask:valid_min = 1b ;
                mask:valid_max = 31b ;
        byte sea_ice_fraction(time, lat, lon) ;
                sea_ice_fraction:standard_name = "sea_ice_area_fraction" ;
                sea_ice_fraction:long_name = "sea ice area fraction" ;
                sea_ice_fraction:units = "1" ;
                sea_ice_fraction:scale_factor = 0.01f ;
                sea_ice_fraction:add_offset = 0.f ;
                sea_ice_fraction:_FillValue = -128b ;
                sea_ice_fraction:comment = "" ;
                sea_ice_fraction:source = "OSI-SAF sea ice concentration" ;
                sea_ice_fraction:valid_min = 0b ;
                sea_ice_fraction:valid_max = 100b ;
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// global attributes:

:Conventions = "CF-1.4, Unidata Observation Dataset v1.0" ;

:title = "Arctic Sea and Ice Surface Temperature, L4, 5km daily (DMI-ARC-SEAICE_TEMP-L4-NRT-OBS)" ;

:summary = "DMI Sea and Ice Surface Temperature analysis created using multisensor satellite surface temperature (ST) observations" ;

:references = "Høyer, J. L. and She, J., Optimal interpolation of sea surface temperature for the North Sea and Baltic Sea, J. Mar. Sys., Vol 65, 1-4, pp. 176-189, 2007" ;

:institution = "Danish Meteorological Institute, DMI" ;

:history = "Version 1.0" ;

:license = "these data are available free of charge under the GMES data policy" ;

:id = "DMI-L4UHskin-ARC_IST-DMI_OI" ;

:naming_authority = "org.ghrsst" ;

:product_version = "Version 1.0" ;

:uuid = "0872317e-5950-4281-a10b-b766d731a130" ;

:gds_version_id = "2.0" ;

:netcdf_version_id = "netcdf-3.6.2 of Sep 10 2008 10:39:34" ;

:date_created = "20250508T110159Z" ;

:file_quality_level = 3 ;

:spatial_resolution = "0.05 degrees" ;

:start_time = "20250507T000000Z" ;

:time_coverage_start = "20250507T000000Z" ;

:stop_time = "20250508T000000Z" ;

:time_coverage_end = "20250508T000000Z" ;

:westernmost_longitude = -179.975f ;

:easternmost_longitude = 179.975f ;

:southernmost_latitude = 58.f ;

:northernmost_latitude = 89.95f ;

:source = "EUR-L2P-AVHRR_METOP_A, OSI-SAF ice concentration" ;

:platform = "MetOpA" ;

:sensor = "AVHRR" ;

:Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;

:metadata_link = "http://data.nodc.noaa.gov/NESDIS_DataCenters/metadata/index.html" ;

:keywords = "Oceans > Ocean Temperature > Sea Surface Temperature" ;

:keywords_vocabulary = "NASA Global Change Master Directory (GCMD) Science Keywords" ;

:standard_name_vocabulary = "NetCDF Climate and forecast (CF) Metadata Convention" ;

:geospatial_lat_units = "degrees north" ;

:geospatial_lon_units = "degrees east" ;

:geospatial_lat_resolution = 0.05f ;

:geospatial_lon_resolution = 0.05f ;

:acknowledgment = "Please acknowledge the use of these data with the following statement: These data were provided by DMI and the MyOcean regional data assembly centre" ;

:creator_name = "DMI" ;

:creator_email = "myocean_po@dmi.dk" ;

:creator_url = "http://ocean.dmi.dk/satellite/" ;

:project = "Copernicus Marine Environmental Monitoring Service (CMEMS) SI-TAC" ;

:publisher_name = "MyOcean and GHRSST project office" ;

:publisher_url = "http://www.myocean.eu.org/ and http://www.ghrsst.org" ;

:publisher_email = "servicedesk@myocean.eu.org , ghrsst-po@nceo.ac.uk" ;

:processing_level = "L4" ;

:cdm_data_type = "grid" ;

:comment = "IN NO EVENT SHALL DMI OR ITS REPRESENTATIVES BE LIABLE FOR ANY DAMAGES WHATSOEVER INCLUDING, WITHOUT LIMITATION, SPECIAL, INDIRECT, INCIDENTAL OR CONSEQUENTIAL DAMAGES OR DAMAGES FOR LOSS OF BUSINESS PROFITS OR SAVINGS, BUSINESS INTERRUPTION, LOSS OF BUSINESS INFORMATION OR OTHER PECUNIARY LOSS

ARISING OUT OF THE USE OF OR THE BUSINESS INTERRUPTION, LOSS OF BUSINESS INFORMATION OR OTHER PECUNIARY LOSS ARISING OUT OF THE USE OF OR THE INABILITY TO USE THIS DMI PRODUCT, EVEN IF DMI HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THIS LIMITATION SHALL APPLY TO CLAIMS OF PERSONAL INJURY TO THE EXTENT PERMITTED BY LAW. SOME COUNTRIES OR STATES DO NOT ALLOW THE EXCLUSION OR LIMITATION OF LIABILITY FOR CONSEQUENTIAL, SPECIAL, INDIRECT, INCIDENTAL DAMAGES AND, ACCORDINGLY, SOME PORTIONS OF THESE LIMITATIONS MAY NOT APPLY TO YOU. BY USING THIS PRODUCT, YOU HAVE ACCEPTED THAT THE ABOVE LIMITATIONS OR THE MAXIMUM LEGALLY APPLICABLE SUBSET OF THESE LIMITATIONS APPLY TO YOUR USE OF THIS PRODUCT. WARNING Some applications are unable to properly handle signed byte values. If values are encountered > 127, please subtract 256 from this reported value" ;

}